

C++ // <iostream>

// include <iostream>

using namespace std;

void
int main()

std::cout <<

2x+4=4

data type

// include <iostream>

using namespace std;

error

int main()

int a=254, c;

c=a+b;

cout << c;

cout << "a+b=" << a+b;

return;

C++

function

a=10

3+4=7

data type

3+4

C++

data type
int
"size"

* Include <iostream>
using namespace std;

error

```
int main() {
    int a = 3, b = 5, c;
    c = a + b;
```

تقليل
المتغيرات
(3, 5, 7)

```
c = a + b;
cout << c;
```

```
cout << c << " = " << a << " + " << b << endl;
cout << "c = " << a + b;
```

return;

}

$$2x + 4 = 4$$

C++

Answer

function

$$a + 0 = 1$$

$$3 + 4 = 7$$

data type variable

$$c = 3 + 4$$

$$c = 7$$

JAVA

22/6/2025

datatype

$$y = \sqrt[3]{x^2 + 7}$$

```
#include <iostream>
# " <math> or <math.h>
using namespace std;
```

```
int main() {
```

```
float x, y;
```

```
cin >> x;
```

```
// y = sqrt(x*x+7);
```

```
// y = cbrt(x*x+7);
```

```
y = Pow(x*x+7, 1/7);
```

```
cout << y;
```

```
return;
```

$$y = \sqrt[8]{3x^4 + 2m}$$

header

```
int main() {
```

```
float y, x, m;
```

```
cin >> x, m;
```

```
y = Pow(Pow(x, 4) + 2*m, 1/8);
```

```
cout << y;
```

```
return;
```

datatype

$$m = \log \left(\frac{\cos^3 x + y^{12}}{\sin 3x^3 + n} \right)$$

```
#include <iostream>
# " <math> or <math.h>
using namespace std;
```

```
int main() {
```

```
float m, n, x, y;
```

```
cin >> x, y, n;
```

```
m = log ( (Pow (cos x, 3) + Pow (y, 12)) /
```

```
Sin Pow (3 * x, 3) + n) ;
```

```
cout << m;
```

```
} return;
```

ava

16/2/25

datatype

write your name:

my name is: _____

write your RegNo:

my RegNo: _____

* header

```
int main() {
```

```
    string name;
```

```
    int RegNo;
```

```
    cout << "Write your Name: \n";
```

```
    (cin >> name);
```

```
    cout << "my name is: " << name << endl;
```

```
    cout << "Write your RegNo: \n";
```

```
    (cin >> RegNo);
```

```
    cout << "my RegNo is: " << RegNo;
```

```
    return 0;
```

```
}
```

$$f = \frac{0^{5M} + \sin^3 x^2}{\log 2M + \sqrt{x}}$$

header of the code
~~xx~~ #include <vector>
~~xx~~ int main() {

string n;
 char m;

cout << "Write your text:";

getline(cin, n);

cout << n;
 cin >> m;

n.push_back(m);
 cout << "your text after char is:" << n;

n.pop_back();
 cout << n;

Haral Sahal

Haral Sahal

Haral Sahal

write your sentences: text
 your sentence after char is: a!
 your sentence after deletion by using pop is: _____

+4

Header

int main() {

bool n1, n2, m;

n1 = true;

n2 = false;

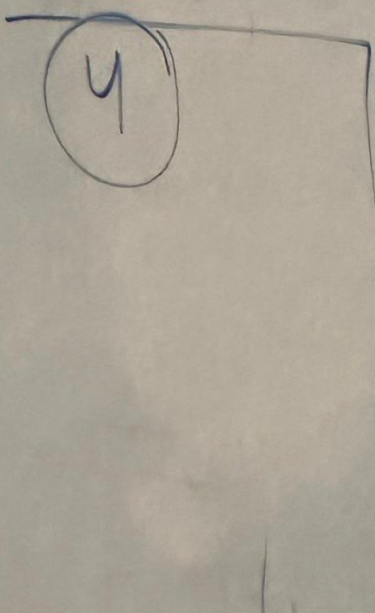
m = 2 * (n1 + n2 + n1);

cout << m;

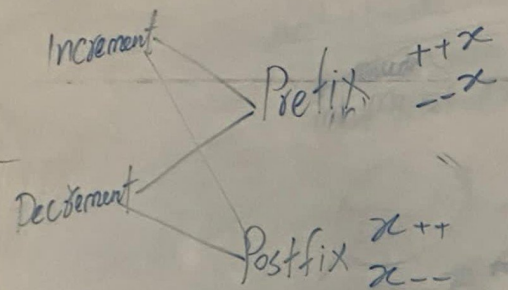
return;

}

8,6



+4



header

```
int main() {
```

```
int x, i;
```

```
x = 4;
```

```
i = ++x; ✓
```

```
cout << "x <=" << i;
```

```
return 0;
```

```
}
```

++x

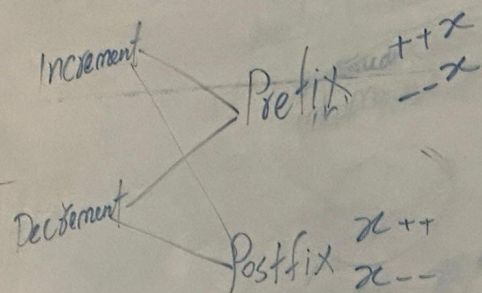
x	4	5
i	5	
5, 5		

x++

x	4	5
i	4	

5, 4

+C



--x

x	32
i	2

2, 2

header

int main {

int x, i;

x = 3;

// i = --x;

i = x--;

cout << x << " " << i;

return;

}

x--

x	32
i	3

2, 3

header

int main

int a, b, c;

a = 3;

b = 5;

c = ++a - b--;

cout << c << ", " << a << ", " << b;

return

}

a 84

b 84

c -1

a 84

b 84

c -1

-1, 4, 4

Increment

Prefix

++x

--x

Decrement

Postfix

x++

x--

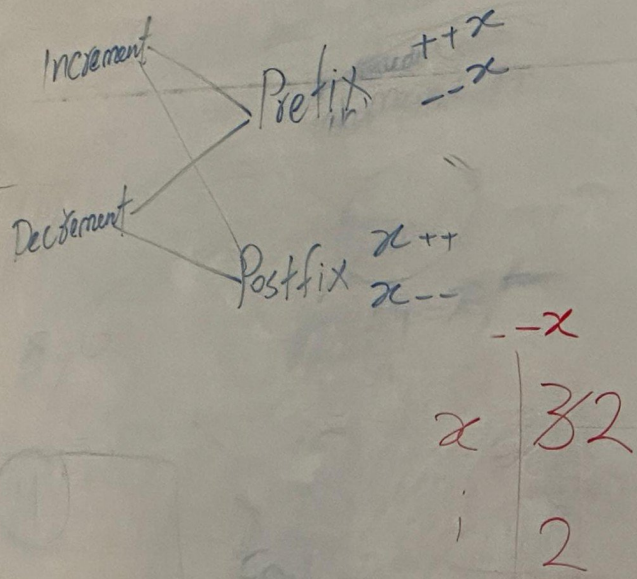
--x

x 82

i 2

2, 2

+C



2, 2

header

int main {

int a, b, c, d;

a = 3;

b = 5;

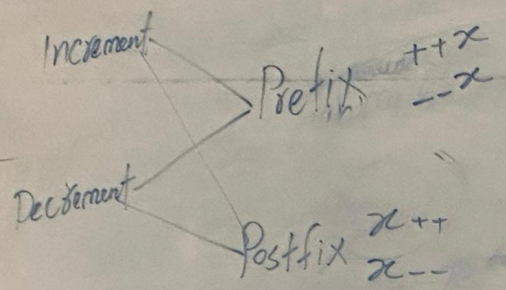
c = 7;

d = ++a * b-- + c++ * 1;

cout << d;

return

a	3	4	27
b	5	4	
c	7	8	
d			



```

int main() {
    int x, y, z;

    x = 4;
    y = 10;
    z = ++x * y-- * 2;
    cout << z << ", " << y;
}
  
```

return;

x	5
y	9
z	100, 9

```

int main() {
    int x, y, z;

    x = 10;
    y = 2;
    f = 3;
    z = ++y * x-- + 5 * f;
}
  
```

return;

x	10
y	3
z	33
f	3

```

int main() {
    int x, y, n, m;

    n = 2;
    m = 4;
    x = 3;
    y = ++n + --x - ++m;
    cout << y;
}
  
```

return;

n	3
m	4
x	2
y	0

3 + 2 = 5

① if (condition)

② if (condition) {
 // body
 }
 else {
 }

③ if (condition) {
 }
 else if (condition) {
 }
 else {
 }

Output

this will execute

header

int main() {

bool b₁, b₂;

b₁ = 0;

b₂ = 1;

if (b₁ ~~||~~ ^{and} b₂ > b₁) cout << "this won't execute";

if (b₁ ~~||~~ ^{and} b₂ >= b₂) cout << "this will execute";

if (b₂ ~~||~~ ^{or} b₁ >= b₁) cout << "this will execute";

if (b₂ || b₁ >= b₁) cout << "this will execute";

return;

① if (condition)

② if (condition) {
 body
}
else {
 body
}

③ if (condition) {
 body
}
else if (condition) {
 body
}
else {
 body
}

header

int main() {

int x;

cin >> x;

① if (x % 2 == 0) {

cout << "this number is even";

else {

cout << "this number is odd";

return 0;

even } 5
odd } 6

5.
 this number is odd
 6
 this number is even

+cc

① if (condition)

② if (condition) {
body

}
else {
}

③ if (condition) {

}
else if (condition) {
}
else {
}

header

int main() {

int h1, h2, m1, m2, s1, s2;
totalSecond = 0;
cout << "enter your time in";
cout << "enter by hours, min, Second";

Cin >> _____;

Cin >> _____;

if (h1 > h2) {

even }
odd }

enter your time
by hours, min, Second
h : m : s
Second
Total in Second;

8
7

int main() {

int age;

cin >> age;

if (age > 14) {

if (age > 18) {

cout << "adult";

else {

cout << "teen";

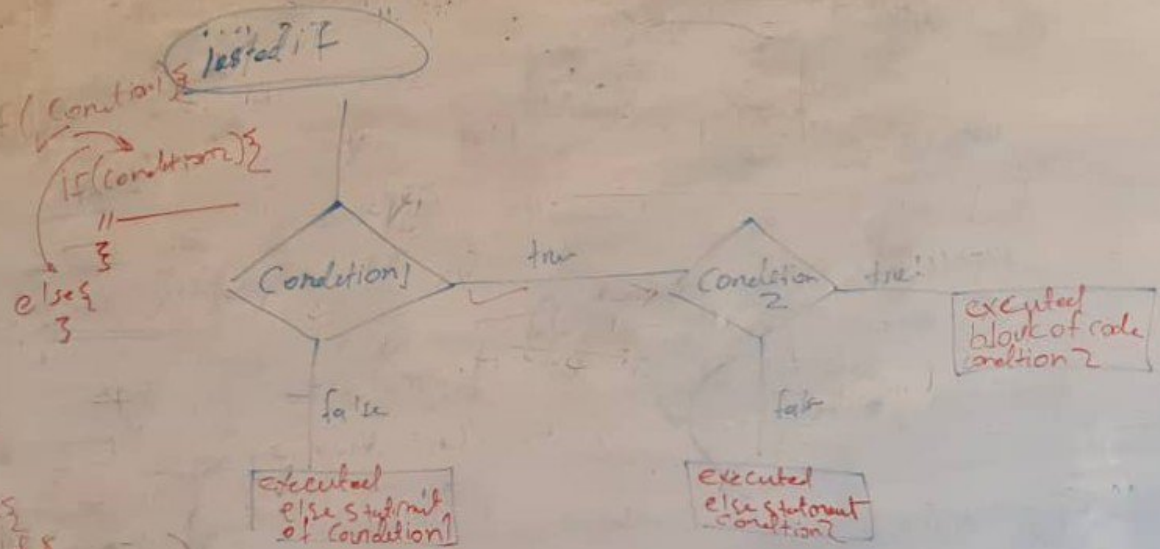
else {

if (age < 0) {

cout << "invalid";

return 0;

8
7



int min1;

int a, b, c;
cin >> a >> b >> c;

if (a > b) {

if (a > c) {

cout << a << endl;

else {

cout << c << endl;

else {

if (b > c) {

cout << b << endl;

else {

12 | 14 | 6

8/7

```

int main() {
    int age;
    cin >> age;
    if (age >= 18) {
        cout << "You can vote";
    }
    else {
        cout << "You can't vote";
    }
}

```

return;

You can vote

int main() {

```

    int mark;
    cin >> mark;
    if (mark > 100 || mark < 0) {
        cout << "out of Rang";
    }
}

```

case 2
into "you got f" }

return;

```

    else if (mark >= 90) {
        cout << "your Grade is A";
    }
    else if (mark >= 80 || mark < 90) {
        cout << "your Grade is B";
    }
    else if (mark >= 70 || mark < 80) {
        cout << "your Grade is C";
    }
    else if (mark >= 60 || mark < 70) {
        cout << "your Grade is D";
    }
}

```

out of Rang

your Grade is A
" " " " " B
" " " " " C
" " " " " D

```
int main() {
```

int age;

Cin $\rightarrow$ age

$$if (age \geq 18) \{$$

Confess " You can love

3

else {

Conte "You Can't Vote".

3

неформе

7

int main() {

```
int weight;
```

Int weight
f but shipping cost;

Grav Weight;

-if (weight ⁷⁰ < 5) {

Shipping-Cost = 0.01

3

```
else if (weight > 50 || weight < 50) {
```

shipping cost = 10

3

else 3

Shipping-Cost = 30;

Cont. "Shipping cost is" "Shipping cost"

Shipping cost is 0.10

10

Your grade is A

11. 11. 11

1/11

8
7

Write program that count the difference between two times by using C++

```

1
hours = totalSeconds / 3600;
minutes = (totalSeconds % 3600) / 60;
seconds = totalSeconds % 60;
cout << "enter the first time\n";
cout << "enter hours, minutes and seconds\n";
totalSeconds = seconds + (minutes * 60) + (hours * 3600);
cout << "enter the second time\n";

```

cout << "enter hours, minutes and seconds\n";

Cin >> h1 >> m1 >> s1;

h1 > h2
4 30 20

h2 > h1
5 20 20

totalSeconds = 84400 - ((h1 * 3600) + (m1 * 60) + s1);

totalSeconds = ((h2 * 3600) + (m2 * 60) + s2) - totalSeconds;

else {

totalSeconds = ((h2 * 3600) + (m2 * 60) + s2) - ((h1 * 3600) + (m1 * 60) + s1);

cout << "total of seconds : ";
return 0;

```

Enter first time
Enter hours, minutes and seconds: ---
Enter second time
Enter hours, minutes and seconds: ---
Total of seconds: ---
hours:
minutes:
seconds:

```

if (totalSeconds < 0)

if (totalSeconds < 0)

else {

cout << "total of seconds : ";

return 0;

loops

for

for (initialization, condition, update)

for [x] // block of code
{ x }

for (var : Array)

do-while

initialization;

{ do {
update // block of code
function
}

} while (condition);

while

initialization;

while (condition) {

// block of code
update, count
}

loops

for
int main() {

// int a;

for(int a = 1; a <= 5; a++) {
 cout << a << endl; }

return;

do-while

int main() {
 int a = 1;

do {
 cout << a << endl;
 a++;

} while (a <= 5);

return;

while

int main() {

int x = 1;

while (x <= 5) {

 cout << x << endl;

 x++;

}

return;

}

loops

Write program that sum

five of numbers by using for loop

```
#include <iostream>
```

```
using namespace std;
```

```
int main() {
```

```
    int x, sum;
```

```
    sum = 0;
```

```
    for (x = 1; x <= 5; x++) {
```

```
        Sum += x;
```

```
    } cout << sum << endl;
```

```
    return 0;
```

Sum = Sum + x

1 = 0 + 1

3 1 + 2

6 3 + 3

10 6 + 4

15 10 + 5

1 2 3 4 5

1

3

6

10

15

1

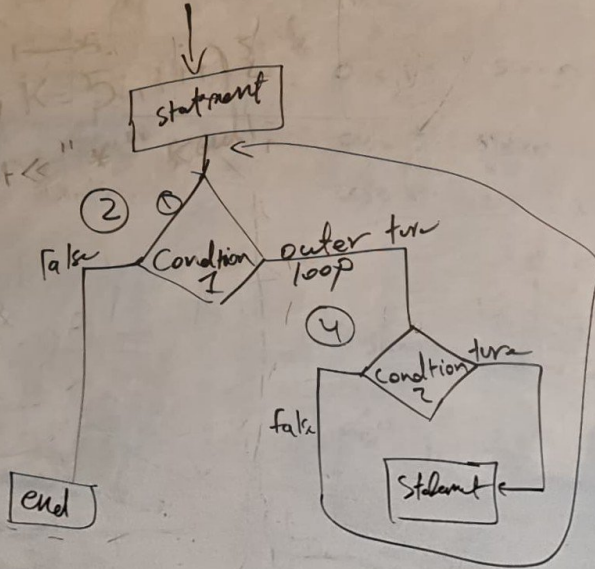
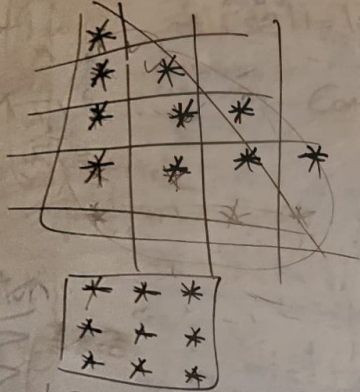
3

6

10

15

nested loop



header

```

int main() {
    for(int a=1; a<=5; a++) {
        for(int b=1; b<=a; b++) {
            cout << "*" ;
        }
    }
}
    
```



```

    cout << endl;
}
    
```

1 2
1 2 3
1 2 3 4
1 2 3 4 5

(loop)

	1			
	1	2		
	1	2	3	
*	1	2	3	4
*				
*				
*				
*				

gs ← header

```
int main {  
    for(int a=1; a<=4; a++) {  
        for(int b=1; b<=a; b++) {  
            cout<<b; }  
        cout<<"\n";  
    }  
    return 0;  
}
```

1	2		
1	2	3	
1	2	3	4
1	2	3	4
1	2	3	
1	2		

(loop)

	1			
	1	2		
	1	2	3	
*	1	2	3	4
*				
*				

1
2 2
3 3 3
4 4 4 4
5 5 5 5 5

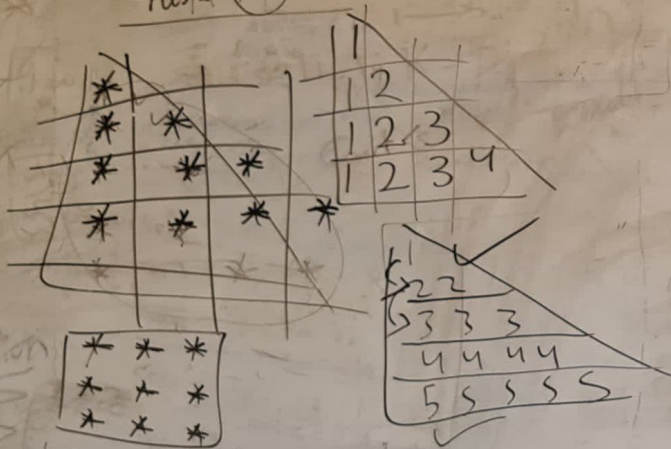
es ← header

int main() {
 for(int a=5; a>=1; a--) {
 for(int b=1; b<=a; b++) {
 cout << b/a; }
 }
 return 0;
}

cout << endl; }
return 0;
}

	b	a
✓	1	5
✓	2	5
✓	3	5
✓	4	5
✓	5	5
✓	1	4
✓	2	4
✓	3	4
✓	4	4
✓	5	4
✓	1	3
✓	2	3
✓	3	3
✓	4	3
✓	5	3
✓	1	2
✓	2	2
✓	3	2
✓	4	2
✓	5	2
✓	1	1
✓	2	1
✓	3	1
✓	4	1
✓	5	1

nested loop



js ← header

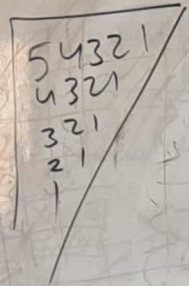
```
int main {
    for(int a=1; a<=5; a++) {
        for(int b=1; b<=a; b++) {
```

cout<<a; }

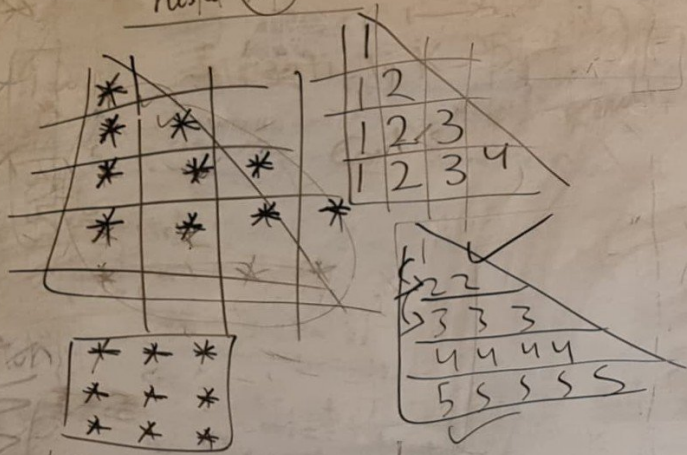
cout<<"endl"; }

return;

}



nested loop



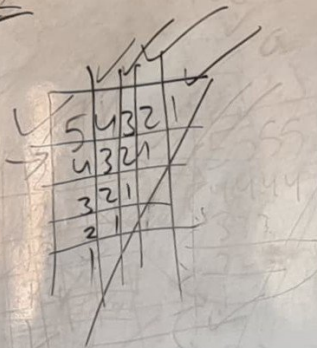
Es ← header

```
int main() {
    for (int a = 5; a = 1; a--) {
        for (int b = 1; b = 32; b++) {
            cout << a;
        }
    }
}
```

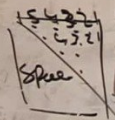
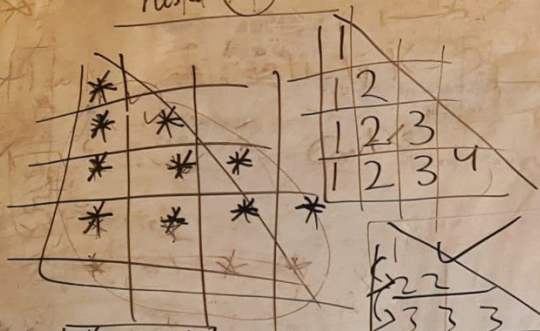
→ cout << endl; }

return;

}



nested loop



Es ← header

Week 1:

day 1
" 2
" 3
" 4
" 5

Week 2:

day 1
" 2
" 3
" 4
" 5
" 6
" 7

Week 3:

day 2
" 4
" 6

Week 4:

day 1
" 3
" 7

con 1

```
for(int a = 1; a <= 4; a++) {
    cout << "week" << a << endl;
    for(int b = 1; b <= 7; b++) {
        if(b % 2 == 0) f(b);
        cout << "day " << b << endl;
    }
}
```

```
if(b % 2 == 0) f(b);
cout << "day " << b << endl;
for(int b = 1; b <= 7; b++) {
    if(b == 5) {
        break;
    }
}
```

header

```
int main() {
    int row;
    cin >> row;
    for (int a = 1; a <= row; a++) {
```

```
        for (int s = a; s <= row; s++) {
```

```
            cout << "  ";
```

```
        } for (int b = 1; b <= a; b++) {
```

```
            cout << " * ";
```

```
        } cout << endl;
```

```
    } return 0;
```



+ , / , - , *
operator
 n1, n2

output
4 + 3 = 7

* include <iostream>
 using namespace std;

int main() {
 char oper;

n1 n2
 4 3

int n1, n2;

4 + 3 = 7

→ cin >> oper; +
 cin >> n1 >> n2;

switch (oper) {

→ Case '+' :

cout << n1 << "+" << n2 << " = " << n1 + n2;
 break;

Case '-' :

cout << n1 << "-" << n2 << " = " << n1 - n2;
 break;

switch (expression) {

Case Constant 1

break;

Case Constant 2 :

Case '*' :
 cout << n1 << "*" << n2 << " = " << n1 * n2;
 break;

Case '/' :
 cout << n1 << "/" << n2 << " = " << n1 / n2;
 break;

default :
 cout << " Invalid " ;

}
 return 0;
 }

header

```
int main() {
    int row, col;
    cin >> row;
    for (int a = 1; a <= row; a++) {
```

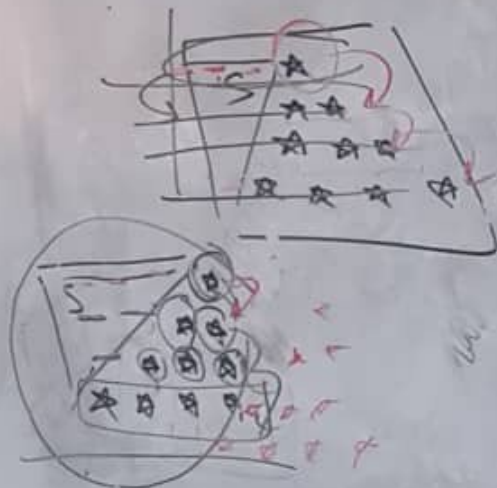
```
        for (int s = 1; s <= row - a; s++) {
```

```
            cout << " ";
```

```
            for (int b = 1; b <= a; b++) {
                cout << "*";
```

```
            }
            cout << endl;
```

return 0;



```
#include <iostream>
using namespace std;
```

```
int main() {
```

```
    int month;
```

```
    cin >> month;
```

```
    switch (month) {
```

```
        case 1: cout << "Jan";  
                break;
```

```
        case 2: cout << "Feb";  
                break;
```

```
        case 3: cout << "March";  
                break;
```

```
        case 4: cout << "April";  
                break;
```

```
        case 5: cout << "May";  
                break;
```

switch

12

```
{ default:  
  cout << "error";  
  break;
```

```
switch (expression) {
```

```
    case constant 1:
```

```
        break;
```

```
    case constant 2:
```

```
        break;
```

```
        ...
```

```
    default:
```

```
        break;
```

```
}
```

MENU

RUN

Coding C++

Auto saved at ۲۳:۱۵:۰۹

```
1 #include <iostream>
2 using namespace std;
3 int main () {
4     int mark;
5     cin >> mark;
6     if ( mark < 1 || mark > 100 ) {
7         cout << " Error " ;
8     }
9     switch (mark / 10 ) {
10    case 10:
11    case 9:
12        cout << "A" ;
13        break;
14    case 8:
15        cout << "B" ;
16        break;
17    case 7:
18        cout << "C" ;
19        break;
20    case 6:
21        cout << "D" ;
22        break;
23    default:
24        cout << "F" ;
25    }
26    return 0 ;
27 }
```

Array

2Dim.

multi...

one Dim.

[. . . .]

[]
[]
[]

[: : :]
[: : :]

datatype

name of Array

[] ;

index

0 1 2 3 4

a[0] a[1] a[2] a[3] a[4]

n-1

n

5

Array

One Dim.

[. . . .]

[]

data type

name of Array

[]

header

int main() {

int n;

int arr[n];

cin >> n;

cout << "the numbers are:";

for (int a = 0; a < n; a++) {

cin >> arr[a];

cout << arr[a]; }

24567 return 0; }

24567

multi...

2456

{the numbers are: 2456}

(n) 5

arr[n]

Array

header

one Dim.

the Sum of the number are 456
the averg is 91

int main()

double matrix[5], sum=0.0, averg;

for(int i=0; i<5; i++) {
cin >> matrix[i];
sum += matrix[i];
}

cout << "the Sum of number are" << sum << endl;
averg = sum / 5;

cout << "the averge is" << averg;

0	1	2	3

matrix[0] [1]

0 1 2 3 4
100, 90, 90, 98, 78

Sum = sum + matrix[i]
0 + 100

100 = 100 + 90
190 = 190 + 90

280 = 280 + 98
378 = 378 + 78

Sum = 456
averg = 456 / 5 = 91.2

return 0;

Array

header

Program

one Dim.
output

Process

int main()

double matrix [5], sum = 0.0, avg;

for (int i = 0; i < 5; i++) {
 $\frac{0+1+2+3+4}{5} = \text{sum}$
 avg = sum / 5;

cin >> matrix[i];

sum += matrix[i];

the sum of the number are 456
 the avg is 91

matrix [5]
 sum 0.0
 avg

0 1 2 3 4
 100 90 90 98 78

sum = sum + matrix[i]
 0 + 100

100 = 100 + 90
 190 = 190 + 90

280 = 280 + 98
 378 = 378 + 78

cout << "the sum of number are" << sum << endl;

avg = sum / 5;

cout << "the average is" << avg;

0 1 2 3

--	--	--	--

matrix[0] [1]

Array

```

int n;
double matrix[n], min, max;
cin >> n;
for (int a=0; a<n; a++) {
    cin >> matrix[a];

```

m[0]	m[1]	m[2]	m[3]	m[4]
7	12	13	15	100

Max = 7

```

    }
    max = matrix[0];
    for (int a=0; a<n; a++) {
        if (max < matrix[a]) {
            max = matrix[a];
        }
    }

```

matrix[0] = 13 - 100

7 12 13 15 100

the largest number is 100
the minimum " is 7

```

min = matrix[0];
for (int a=0; a<n; a++) {
    if (min > matrix[a]) {

```

min = matrix[a];

} the largest number is

cout << max << endl;

cout << min << endl;

Array

```
int n;
double matrix[n], min, max;
cin >> n;
for (int a=0; a<n; a++){
    cin >> matrix[a];
}
```

m[0]	m[1]	m[2]	m[3]	m[4]
7	12	13	15	100

Max = 7

```
max = matrix[0];
for (int a=0; a<n; a++){
    if (max < matrix[a]){
        max = matrix[a];
    }
}
```

matrix[0] = 7

7 12 13 15 100

the largest number is 100

the minimum " is 7

```
min = matrix[0];
for (int a=0; a<n; a++){
    if (min > matrix[a]){
        min = matrix[a];
    }
}
```

Min = matrix[0]

cout << "the largest number is" << max << endl;

cout << "the minimum number is" << min << endl;

Array

Hend, Sara, Salem, Ahmed, Mona

header

```
int main() {
```

```
    string name = {"Hend", "Sara", "Salem", "Ahmed", "Mona"};
```

```
    for (int i = 0; i < 5; i++) {
```

```
        cout << name[i] << endl;
```

```
    }
    cout << name[0] << endl;
```

Hend Sara Salem

H

S

S

A

M

H

S

S

A

M

2D Array

header		
00	01	02
10	11	12

```
int main() {  
    int arr[2][3] = { {1, 2, 3}, {4, 5, 6} };  
    for (int i = 0; i < 2; i++) {  
        for (int j = 0; j < 3; j++) {
```

1 2 3
4 5 6
v v

```
        cout << arr[i][j] << " ";  
    } cout << endl;  
    return 0;  
}
```

2D Array

```

if (max < matrix[a][b]) {
    max = matrix[a][b];
}

```

```

min = matrix[0][0];
for (int a = 0; a < R; a++) {

```

```

    for (int b = 0; b < C; b++) {

```

```

        if (min > matrix[a][b]) {

```

```

            min = matrix[a][b];
        }

```

cout << "The max is" << max << endl;

cout << "The min is" << min << endl;

header

```

int main() {
    int R, C;
    double matrix[R][C];
    cin >> R >> C;

```

4	7	40
5	10	22
10	70	100

```

    for (int i = 0; i < R; i++) {

```

```

        for (int j = 0; j < C; j++) {

```

```

            cin >> matrix[i][j];
        }
    }

```

```

    max = matrix[0][0];
    for (int a = 0; a < R; a++) {

```

```

        for (int b = 0; b < C; b++) {

```

8	7	1	2	2
0	0	1	1	2
0	0	1	0	2